Spacecraft telecommunications equations

Equations and assumptions in `main.py`

This reference records the implementation inspected on 14 October 2026. Table 1 covers all 22 calculation functions in `main.py`, including helpers that are not used by its default execution. Running that file calls `earth_model.calculate_earth()`; Table 2 therefore includes the additional equations needed to assemble its printed link budgets. The equations describe the code as written; differences from the lecture slides are explicitly identified.

Sources. S3 and S4 refer to the two supplied AE2111-II Telecommunications lecture PDFs. Slide numbers equal the PDFs' one-based page numbers. "Code" means an implemented expression without an explicit matching equation in those slides. "Derived" means an algebraic or geometric step used in the code. MW refers to the MathWorks source listed at the end.

Notation and units. Unless stated otherwise, R and h are body radius and orbital altitude in km, μ is in km^3/s^2, d is range, D is antenna diameter in m, f is frequency, and λ is wavelength in m. α is station elevation and ψ is the angle at the Sun between the Earth–Sun and body–Sun lines; trigonometric arguments are in radians. β is the full cross-track field of view in degrees, δ is pixel angle in arcminutes, b is bits per pixel, and u is duty cycle in percent. R_i and R_c denote information and channel bit rates in bit/s. $\ell_t, \ell_r \in (0, 1]$ are linear transmitter and receiver cable transmission factors, distinct from positive attenuation magnitudes L in dB. Temperatures are in K. $\gamma = E_b/N_0$ is linear; $\Gamma = 10 \log_{10} \gamma$ is in dB. Logarithms of dimensional quantities use their numerical values in the stated reference units, as in the code and slides.

Table 1: Equations implemented in `main.py`.

Quantity / function	Implemented equation	Source and conditions
Circular orbital period `orbital_period`	$T_{\text{orb}} = 2\pi\sqrt{\dfrac{(R+h)^3}{\mu}}$	*Code; circular Keplerian orbit.* Output in s. Used by the ground-track helper, which the default budget does not call.
Circular orbital speed `orbital_speed`	$v_{\text{orb}} = 10^3\sqrt{\dfrac{\mu}{R+h}}$	*Code; circular Keplerian orbit.* The 10^3 converts km/s to m/s. This is the scan speed used for the payload.
Nadir ground-track speed `ground_track_speed`	$v_g = \dfrac{2\pi R\, 10^3}{T_{\text{orb}}} = v_{\text{orb}}\dfrac{R}{R+h}$	*Derived; non-rotating spherical body.* Available helper; not used by the default payload calculation.
Angular conversion and nadir pixel size `ground_pixel_size`	$\theta_p = \dfrac{\delta\pi}{180 \cdot 60}$, $s_p = 2h\, 10^3 \tan(\theta_p/2)$	*Code; local flat-surface geometry.* Output s_p in m; square angular pixels.
Pixels per line `pixels_per_line`	$N_x = \left\lceil \dfrac{60\beta}{\delta} \right\rceil$	*Code.* Counts angular pixels across the full field of view; rounds upward to an integer.
Active payload rate `payload_data_rate`	$f_{\text{line}} = \dfrac{v_{\text{orb}}}{s_p}$, $R_{\text{payload}} = N_x f_{\text{line}} b$	*Code; nadir pushbroom camera.* One row per along-track pixel. Orbital speed approximates ground scan speed; no compression or packet overhead.
Daily raw data `daily_data_generated`	$V_{\text{day}} = R_{\text{payload}}\, 86400\, \dfrac{u}{100}$	*Derived from rate × time.* Output in bits; $0 \le u \le 100$.
Required downlink information rate `required_downlink_rate`	$R_{i,\text{DL}} = \dfrac{V_{\text{day}}}{3600\, t_{\text{DL}}}$	*Derived from data / contact time.* t_{DL} in hr/day. This is an average over the assumed daily transmission time, before coding overhead.
Atmospheric attenuation `atmospheric_attenuation_noise`	$L_a(f) = \begin{cases} 0.15, & 0 < f \le 5 \cdot 10^9, \\ 0.20, & 5 \cdot 10^9 < f \le 10^{10}, \\ 0.75, & f > 10^{10} \end{cases}$	*Code approximation; S4, slides 15–16 provide context.* f in Hz; output in dB. Fixed at an assumed 20° elevation. The slide equation $A = A_{90^\circ} / \sin\alpha$ is not evaluated.

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Table 1 continued

Quantity / function	Implemented equation	Source and conditions
Linear to decibels `n_dB`	$x_{\mathrm{dB}} = 10\log_{10}(x/1)$	*S3, slide 14.* $x > 0$. For power entered in W, the result is dBW; for a dimensionless factor, it is dB.
Decibels to linear `dB_n`	$x = 10^{x_{\mathrm{dB}}/10}$	*Inverse of S3, slide 14.* Unit reference is 1. Available helper; not used by the default budget.
Parabolic antenna gain `antenna_gain`	$G_{\mathrm{dBi}} = 10\log_{10}\left[\eta\left(\frac{\pi D}{\lambda}\right)^2\right]$	*S4, slide 24.* η is the supplied efficiency. The active adapter assumes $\eta = 0.6$; the example on slide 34 uses 0.55.
Free-space attenuation `freespaceloss`	$L_{\mathrm{FS}} = 20\log_{10}\left(\frac{4\pi d}{\lambda}\right)$	*S3, slide 30; S4, slides 8 and 10.* d, λ in m. Positive attenuation is subtracted from the budget. The function rejects negative calculated attenuation; the model assumes far-field free-space propagation.
Beamwidth and pointing attenuation `pointing_angle_loss`	$\theta_{3\mathrm{dB}} = \frac{21}{f_{\mathrm{GHz}} D},$ $L_p = 12\left(\frac{\theta_{\mathrm{off}}}{\theta_{3\mathrm{dB}}}\right)^2$	*Beamwidth: S4, slide 34.* *Pointing-loss fit: code.* Angles in degrees. The code treats $21/(fD)$ as the full half-power beamwidth; the quadratic loss fit is an approximation. Evaluate for both antennas.
Available energy-per-bit ratio `energy_per_bit`	$\Gamma_{\mathrm{avail}} = \mathrm{EIRP} - L_{\mathrm{FS}} - L_x + (G/T) - 10\log_{10}(k_B R_i)$	*S4, slide 59.* EIRP in dBW, G/T in dB/K, losses in dB, $R_i > 0$ in bit/s. $k_B = 1.380649 \times 10^{-23}$ J/K. Uses information-bit energy.
Earth slant range `distance_earth_orbit`	$d_E = R_E\left[\sqrt{\left(\frac{R_E + h}{R_E}\right)^2 - \cos^2\alpha} - \sin\alpha\right]$	*S4, slide 10.* Distances in km; α in radians. The active Earth case uses the minimum elevation 20°.
Lunar tangent-path range `distance_other_orbit`, Moon branch	$d_M = d_E(h_M, \alpha),$ $d = \sqrt{(R_M + h)^2 - R_M^2} + \sqrt{d_M^2 - R_M^2}$	*Code; tangent geometry.* $h_M = 384400$ km is passed to the Earth-altitude formula; $R_M = 1740$ km. This helper is not called by the current adapter.
Planetary tangent-path range `distance_other_orbit`, Mars / Mercury branches	$d_{EP} = \sqrt{a_E^2 + a_P^2 - 2a_E a_P\cos\psi},$ $d = \sqrt{d_{EP}^2 - R_P^2} + \sqrt{(R_P + h)^2 - R_P^2} - R_E\sin\alpha$	*Code; cosine rule and tangent geometry.* Distances in km, angles in radians, $a_E = 1$ AU. Only Mars and Mercury are supported here. The current adapter uses a different range approximation; see Table 2.
QAM parameters and inverse Gaussian tail `required_eb_n0`	$m = \log_2 M,$ $A_M = \frac{4}{m}\left(1 - \frac{1}{\sqrt{M}}\right),$ $q = Q^{-1}(p_b/A_M) = -\Phi^{-1}(p_b/A_M)$	*Code; MW, square-QAM analysis.* $M \in \{4, 64\}$; target BER $0 < p_b \le 10^{-3}$. $Q(x) = 1 - \Phi(x)$, with Φ the standard-normal CDF. The leading BER term is $p_b \simeq A_M Q(\sqrt{3m\gamma/(M-1)})$.

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Table 1 continued

Quantity / function	Implemented equation	Source and conditions
Required uncoded and coded E_b/N_0 `required_eb_n0` (continued)	$\gamma_{\mathrm{req}} = \frac{M-1}{3m}\, q^2,$ $\Gamma_{\mathrm{req}} = 10\log_{10}\gamma_{\mathrm{req}} - G_c,$ $G_c = \begin{cases} 0, & \text{uncoded,} \\ 10, & \text{coded or LDPC} \end{cases}$	*QAM inversion: code / MW. Coding gain: S4, slide 72.* For 4-QAM, $p_b = Q(\sqrt{2\gamma})$ is exact in the assumed channel. For 64-QAM, the low-BER approximation is used. A fixed gain replaces a code-specific BER model.
Channel bit rate `transmitted_bit_rate`	$R_c = R_i/r_c,$ $r_c = \begin{cases} 1, & \text{uncoded,} \\ 3/4, & \text{coded or LDPC} \end{cases}$	*S4, slide 72; derived rate conversion.* Coding adds $33\frac{1}{3}\%$ channel bits for rate 3/4; generated payload data remain unchanged.
Noise factor and receiver noise `receiver_noise_and_gt`	$F = 10^{NF/10},$ $T_{\mathrm{rx}} = T_0(F-1)$	*S3, slide 37; dB conversion from slide 14.* NF in dB. T_{rx} is referred to the receiver's own input.
Antenna-referred cable and receiver noise `receiver_noise_and_gt` (continued)	$T_{c,A} = T_c(1/\ell_r - 1),$ $T_{\mathrm{rx},A} = T_{\mathrm{rx}}/\ell_r$	*Cable noise: S3, slide 38. Receiver referral: code.* The receiver contribution is divided by the preceding cable's transmission factor to refer it to the antenna terminals.
System noise and receiving figure of merit `receiver_noise_and_gt` (continued)	$T_s = T_A + T_{c,A} + T_{\mathrm{rx},A},$ $(G/T) = G_{r,\mathrm{dBi}} - 10\log_{10} T_s$	*S3, slides 43–44 and 48, with a code difference.* Slides 43–44 print $T_0(F-1)$ without $1/\ell_r$; the code includes that referral factor. Receive cable loss is already represented in this G/T.
Rain attenuation `rain_attenuation_noise`	$L_r(f) = \begin{cases} 0, & f \le 5\cdot 10^9, \\ 2.86, & 5\cdot 10^9 < f < 12\cdot 10^9 \end{cases}$	*Code approximation; S4, slides 17–20 provide context.* f in Hz; output in dB. The helper raises an error at $f \ge 12$ GHz. No location, rain zone or availability probability is supplied.
Link margin `link_margin`	$\mathcal{M} = \Gamma_{\mathrm{avail}} - \Gamma_{\mathrm{req}}$	*Code; S4, slide 84 discusses design margin.* Output in dB. The slide's suggested 3 dB margin is not an enforced criterion in this function.

Table 2: Additional equations in `earth_model.py` called by the `main.py` entry point.

Quantity / location	Equation	Source and conditions
Uplink frequency `calculate_scenario`	$f_{\mathrm{UL}} = \tau f_{\mathrm{DL}}$	*Code / mission input.* τ is defined as uplink frequency divided by downlink frequency.
Wavelength `_link_budget`	$\lambda = \frac{c}{10^9 f_{\mathrm{GHz}}}$	*Derived from* $c = f\lambda$. $c = 299792458$ m/s. Both links use their own carrier wavelength.
Transmitter cable loss and EIRP `_link_budget`	$L_t = -10\log_{10}\ell_t,$ $\mathrm{EIRP} = 10\log_{10} P_t + G_t - L_t$	*S3, slide 49, using positive loss magnitudes.* P_t in W. Uplink transmitter: ground station; downlink transmitter: spacecraft.
Other propagation / pointing losses `_link_budget`	$L_x = L_a + L_r + L_{p,\mathrm{SC}} + L_{p,\mathrm{GS}}$	*Code; loss aggregation in S4, slide 59.* Includes pointing at both ends. Does not subtract the receive cable factor a second time.

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Table 2 continued

Quantity / location	Equation	Source and conditions
Carrier-to-noise density `_link_budget`	$C/N_0 = \Gamma_{\text{avail}} + 10\log_{10} R_i$ $= \text{EIRP} - L_{\text{FS}} - L_x$ $+ (G/T) - 10\log_{10} k_B$	*Derived from S3, slide 29 and S4, slide 59.* Output in dB-Hz. The second expression remains defined for zero information rate; E_b/N_0 and margin then return `None`.
Symbol rate and ideal minimum passband bandwidth `_link_budget`	$m = \log_2 M,$ $R_s = R_c/m,$ $B_{\min} = R_s$	*Code; S4, slides 54–55 support bits per symbol.* Ideal zero pulse-shaping roll-off. No available channel bandwidth is an input. For a raised-cosine model, nonzero roll-off would increase occupied bandwidth.
Actual remote-body range used by adapter `_range_geometry`	$d_{\text{Moon}} = 384400$ km, $d_{EP} = \sqrt{a_E^2 + a_P^2 - 2a_E a_P \cos\psi}$	*Code; centre-distance approximation.* The cosine-rule expression is used for Mars, Venus and Mercury. If ψ is unset, the adapter uses 180°. These replace the unused tangent-path helper in `main.py`.
Daily decimal terabytes `calculate_scenario`	$V_{\text{TB}} = \frac{V_{\text{day}}}{8 \cdot 10^{12}}$	*Unit conversion.* Eight bits per byte; decimal TB, not binary TiB.

Assumptions and fixed choices. The following table distinguishes modelling choices from mission inputs. Values from `earth_model.py` matter because that file supplies the defaults when `main.py` is run. Code comments claiming an assumption is permitted by the assignment are recorded as code comments; the assignment itself was not supplied here.

Table 3: Assumptions found in the implementation.

Topic	Assumption / selected value	Evidence and consequence
Orbit and body	Circular orbit; spherical, non-rotating body; fixed gravitational parameter and radius.	`main.py` physical-assumptions comment and orbital helpers. No rotation, eccentricity or orbital perturbations enter the payload estimate.
Camera model	Nadir pushbroom imaging; square angular pixels; one row per along-track pixel; the nadir pixel footprint is computed on a local flat surface.	`ground_pixel_size` and `payload_data_rate` docstrings. The angular swath/pixel ratio sets the cross-track pixel count.
Scan velocity	$v_{\text{scan}} = v_{\text{orb}}$, rather than the projected ground-track speed v_g.	Explicit `payload_data_rate` approximation. At the Earth defaults, $v_{\text{orb}}/v_g = (R_E + h)/R_E \simeq 1.129$; rates are about 12.9% higher than with the unused ground-track helper.
Payload volume and contact	Raw data only; no compression or packet/protocol overhead. The assumed daily contact time is fully available to transmit one day's collected data at a constant average rate.	Rate and volume helpers; `calculate_scenario` assumptions. Contact availability and the compatibility of imaging with antenna pointing are not computed. FEC overhead is handled separately.
Elevation and geometry	Minimum station elevation is fixed to 20°. The Earth budget uses the range at that elevation for both directions.	`MINIMUM_ELEVATION_DEG` and `distance_earth_orbit`. Atmospheric losses also assume 20° but do not accept elevation as an argument.

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Table 3 continued

Topic	Assumption / selected value	Evidence and consequence
Antenna properties	Both antennas are modelled as parabolic with $\eta = 0.6$, independent of frequency. The same dish diameters are used for transmitting and receiving.	`ANTENNA_EFFICIENCY` and `_link_budget`. Efficiency is separate from cable transmission factors. Slide 34's example uses $\eta = 0.55$.
Pointing	The selected spacecraft offset is used in both links. The active ground offset is 0.01°. Each loss uses the approximate full beamwidth and quadratic pointing fit in Table 1.	`GROUND_POINTING_OFFSET_DEG= 0.01`. The `ASSUMPTIONS` text says 0° and `dashboard.md` says ideal ground pointing; both disagree with the executed constant.
Receiver thermal model	Spacecraft antenna noise $T_A = 290$ K; ground antenna noise $T_A = 100$ K; $NF = 2$ dB for either receiver; cable physical temperature $T_c = 290$ K; reference temperature $T_0 = 290$ K.	Adapter constants and `receiver_noise_and_gt` defaults. Noise and G/T are referred to the antenna terminals. Additional receiver stages are represented by the single aggregate noise figure.
Atmosphere and rain	Frequency-band constant losses: $L_a = 0.15$, 0.20 or 0.75 dB; $L_r = 0$ or 2.86 dB. Antenna noise temperatures remain fixed.	The two attenuation helpers return attenuation, despite "noise" in their names. No atmospheric/rain emission, weather distribution, rain-zone selection or availability calculation is included.
Radio propagation	Far-field free-space spreading and the listed atmospheric, rain and pointing losses. No extra polarization mismatch, interference, multipath or implementation-loss term is included.	`freespaceloss` and adapter loss sum. These effects have no separate budget term.
Modulation and BER	Coherent Gray-coded square QAM in additive white Gaussian noise (AWGN). 4-QAM uses an exact uncoded BER expression; 64-QAM uses the low-BER nearest-neighbour approximation. The same selection applies to both links.	`required_eb_n0` docstring and adapter default $M = 4$. Default target BER is a mission input, 10^{-6}.
Channel coding	Uncoded: $r_c = 1$, $G_c = 0$ dB. "Coded" and "LDPC": $r_c = 3/4$, $G_c = 10$ dB, independent of modulation and BER.	S4, slide 72 supplies the LDPC example. The implementation does not model a particular code, decoder or measured BER curve. The adapter exposes uncoded and LDPC choices.
Energy and bandwidth conventions	Available and required E_b/N_0 are per information bit. Coding overhead changes channel rate and symbol rate. Minimum passband bandwidth equals symbol rate at ideal zero roll-off.	`_link_budget`. R_i, not R_c, enters `energy_per_bit`; coding gain is applied once. Bandwidth feasibility is not established by margin.
Remote-body helper	The unused `distance_other_orbit` assumes tangent geometry and fixed Earth–Moon / heliocentric distances. Mars and Mercury use a geometric angle at the Sun.	`main.py` constants: Moon 384400 km / radius 1740 km; Mars 228×10^6 km / 3390 km; Mercury 46×10^6 km / 2440.5 km. The Moon distance is treated as an altitude argument in this helper.
Active remote-body adapter	Moon uses mean centre distance. Planets use fixed heliocentric radii: Earth 1 AU, Mars 228×10^6 km, Venus 108×10^6 km, Mercury 46×10^6 km (a fixed perihelion approximation). Unset angle defaults to 180°.	`_range_geometry`; not used in the default Earth case. Spacecraft orbital phase, station position, occultation, conjunction and contact availability are omitted.

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Table 3 continued

Topic	Assumption / selected value	Evidence and consequence
Power and margin	RF transmitter power is used directly. Total spacecraft power only limits RF output; amplifier DC efficiency and other subsystem loads are unspecified. The code reports available minus required E_b/N_0.	Adapter validation and budget. S4, slide 84 suggests at least 3 dB design margin; the implementation does not enforce this allowance.

Default Earth mission inputs (not inferred physical assumptions). These are assigned values in the code, labelled as mission-table inputs. Their original mission table was not among the two supplied lecture PDFs.

Table 4: Default inputs used by the Earth entry point.

Input	Value
Orbital altitude	$h = 820$ km
Camera field of view / angular pixel	$\beta = 45°$ full cross-track; $\delta = 0.01$ arcmin
Bits per pixel / imaging duty cycle	$b = 32$ bits total across channels; $u = 100\%$
Daily downlink time	$t_{\mathrm{DL}} = 0.5$ hr/day
Required uplink information rate	$R_{i,\mathrm{UL}} = 10^8$ bit/s
Spacecraft / ground RF transmitter power	150 W / 400 W
Transmitter / receiver cable transmission	$\ell_t = 0.8$ / $\ell_r = 0.7$
Downlink frequency / turnaround ratio	$f_{\mathrm{DL}} = 2.2$ GHz; $\tau = 221/240$
Spacecraft / ground antenna diameter	$D_{\mathrm{SC}} = 1$ m / $D_{\mathrm{GS}} = 15$ m
Spacecraft pointing offset	0.12°
Coding / required BER	Uncoded / $p_b = 10^{-6}$
Adapter modulation default	4-QAM ($M = 4$, two bits per symbol)
Adapter total spacecraft power input	850 W; only an RF-output upper bound in the present validation

Fixed constants and unused input. `main.py` uses $R_E = 6371$ km, $\mu_E = 398600.435507$ km^3/s^2, 1 AU = 149597870.7 km, $c = 299792458$ m/s, $k_B = 1.380649 \times 10^{-23}$ J/K and 86400 s/day. The speed of light and Boltzmann constant are physical constants, although the code groups them under an assumptions comment. The assigned `signalnoise_ratio=3` dB is unused and does not set the required E_b/N_0, link margin or a design allowance. The opening comment labels the link expression "SNR", but `energy_per_bit` computes E_b/N_0. In linear units, $E_b/N_0 = \mathrm{SNR}\, B/R_i$ (S4, slide 56); equality requires $B = R_i$. That bandwidth equality is not an assumption used by the implemented energy-per-bit calculation. The default execution gets its mission inputs from `earth_model.py`; changing only the matching module-level values in `main.py` does not change those adapter defaults.

Implementation checks. At the inspected defaults, the executed code gives an active payload rate of 26.968 Gbit/s, daily raw data of 291.254 decimal TB, a downlink information rate of 1.294462 Tbit/s and an Earth slant range of 1804.572 km. Uncoded 4-QAM at BER 10^{-6} requires 10.530 dB; uncoded 64-QAM requires approximately 18.777 dB under the implemented approximation. The default 4-QAM margins are 43.249 dB uplink and 0.067 dB downlink, using the executed 0.01° ground pointing offset. The ideal downlink minimum bandwidth is approximately 647.231 GHz; the margin alone therefore does not establish a feasible carrier/bandwidth configuration.

Source details. The supplied PDFs are `AE2111-II_20Lecture_S3_Telecommunications_20-_20Copy-2.pdf` and `AE2111-II_20Lecture_S4_Telecommunications_20-_20Copy-2.pdf`. Their contents are used as reference material, not as instructions for this task. The external source already cited by `required_eb_n0` was checked: MathWorks, Analytical Expressions Used in berawgn Function and Bit Error Rate Analysis App, square-QAM and QPSK sections (accessed 14 October 2026). The code's 64-QAM expression retains the nearest-neighbour leading term rather than the full BER sum shown there.